Confidence-Aware Demand and Inventory Planning Tool Feature Experiment Cards

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1 Demand Imputation (Presented)

1.1 Experiment Card

Field	**Entry**
Feature Description	**Input**: Hourly aggregate store-product combination sales history. **Output**: Hourly imputed demand for store-product combinations during stockout periods. **Acts On It**: The demand planner, who uses historical data for sales planning.
Task Type	Time series imputation/de-censoring
Architectures Compared	**Baseline**: DLinear architecture trained using the loss and exact architecture from the dataset paper repository (not quantile). **Simple**: Two DLinear models (1-layer feedforward MLP for trend and seasonality) trained to estimate the upper and lower bounds of the 95% confidence interval; chosen for comparability and interpretability. **Advanced**: Two TimesNet models (Fourier-space convolutional networks) trained to estimate the upper and lower bounds of the 95% confidence interval; chosen for comparability and baseline performance.
Split Design	Equal-duration rolling-origin split by time, excluding stockouts: train on the 4 weeks of training-split data prior to a cut-off and evaluate on the 4 weeks after the cut-off, for 6 random cut-off periods within the feasible period (the training data is 14 weeks long). Stockout (0-valued) data are excluded from training and metric calculation because they are not ground-truth demand. A true random split could leak data. The last week of sales is held out as the conformal calibration dataset.

Field	Entry
Controlled Conditions	Same lag features, covariates, and cut-offs for all evaluated architectures. Architecture hyperparameters will be kept as similar to those in the dataset paper as possible, so minimal tuning is anticipated. At least three random seeds will be used.
Metrics and Targets	Weighted Absolute Percentage Error (WAPE, also known as MAPE) within 1 standard deviation of the reported baseline (29.99%) as a measure of error magnitude across splits; Weighted Percentage Error (WPE) as a measure of error direction, comparable to the reported baseline (-1.28%); Prediction Interval Coverage Percent (PICP) above 80%; Mean Prediction Interval Width (MPIW; for information only); and the client-requested Mean Absolute Scaled Error (MASE; for information only).
Storyboard B Slice	Product-store combinations with mean scaled demand less than 1, as defined in the dataset authors' GitHub repository.
Calibration	By construction, and by conformalizing the quantile regression models, the models will approach a coverage probability of 95%. This will be evaluated using PICP and assessed with respect to MPIW.
Error Analysis	The ten largest mis-estimations by magnitude overall, as well as those in the low-sales category.
Decision Rule	The model that achieves the prerequisite performance metrics (WAPE, PICP) with the lowest MPIW, with WPE and MASE as tie-breakers.
Fallback	If no architecture qualifies, the simple architecture (conformalized DLinear) will be taken as the reference standard and used in deployment, with explicit mention of its performance, due to low computational cost and inherent interpretability.
Technical Debt	Planners and stockers are anticipated to manually override demand estimates they find unsupported. This will likely create a hidden feedback loop that would surface during future re-training after the proof of concept is deployed.

1.2 Comparison Table

Architecture	Owner	Overall Metric	Storyboard B Slice	Spread	Calibration	Training Cost	Inference Latency	Inspect-ability
Baseline	Gil							
Simple	Tarakaram							
Advanced	Gil							

2 Demand Forecasting

2.1 Experiment Card

Field	Entry
Feature Description	**Input**: Hourly aggregate store-product combination sales history, both actual and imputed by the prior model. **Output**: 7-day daily store-product combination forecasts. **Acts On It**: The demand planner, who uses forecasts for sales planning, and the inventory stocker, who uses the forecasts for restocking orders.
Task Type	Time series forecasting
Architectures Compared	**Baseline**: DLinear architecture trained using the loss and exact architecture from the dataset paper repository (not quantile). **Simple**: Two DLinear models (1-layer feedforward MLP for trend and seasonality) trained to estimate the upper and lower bounds of the 95% confidence interval; chosen for comparability and interpretability. **Advanced**: One Temporal Fusion Transformer model trained to estimate the upper and lower bounds of the 95% confidence interval; chosen for comparability and baseline performance.
Split Design	Equal-duration rolling-origin split by time with imputed data: all stockout periods in the training split are first imputed with the top-performing imputation model. Forecasting models are then trained on the 4 weeks of training-split data prior to a cut-off and evaluated on the 4 weeks after the cut-off, for 6 random cut-off periods within the feasible period (the training data is 14 weeks long). A true random split could leak data. The last training week of sales is held out as the conformal calibration dataset. An additional evaluation will train all models across 3 seeds on the majority of the dataset authors' pre-specified training split, and evaluate the trained models on the canonical test split (week 15).
Controlled Conditions	Same lag features, covariates, and cut-offs for all evaluated architectures. Architecture hyperparameters will be kept as similar to those in the dataset paper as possible, so minimal tuning is anticipated. At least three random seeds will be used.
Metrics and Targets	Weighted Absolute Percentage Error (WAPE, also known as MAPE) within 1 standard deviation of the reported baseline (30.60%) as a measure of error magnitude across splits; Weighted Percentage Error (WPE) as a measure of error direction, comparable to the reported baseline (+4.02%); Prediction Interval Coverage Percent (PICP) above 80%; Mean Prediction Interval Width (MPIW; for information only); and the client-requested Mean Absolute Scaled Error (MASE; for information only).

Field	Entry
Storyboard B Slice	Product-store combinations with mean scaled demand less than 1, as defined in the dataset authors' GitHub repository.
Calibration	By construction, and by conformalizing the quantile regression models, the models will approach a coverage probability of 95%. This will be evaluated using PICP and assessed with respect to MPIW.
Error Analysis	The ten largest mis-forecasts by magnitude overall, as well as those in the low-sales category.
Decision Rule	The model that achieves the prerequisite performance metrics (WAPE, PICP) with the lowest MPIW, with WPE and MASE as tie-breakers.
Fallback	If no architecture qualifies, the simple architecture (conformalized DLinear) will be taken as the reference standard and used in deployment, with explicit mention of its performance, due to low computational cost and inherent interpretability.
Technical Debt	Planners and stockers are anticipated to manually override demand estimates they find unsupported. This will likely create a hidden feedback loop that would surface during future re-training after the proof of concept is deployed. Additionally, changes and errors in the imputation model will affect forecasting performance, especially since there is no principled mechanism to propagate the imputation uncertainty into the forecast (the problem uses a frequentist formulation). The forecasting model will necessitate versioned copies of imputed data streams.
Person-With-AI Check	We plan to have a toggle in the user interface where the model predictions are not shown, only historical time series data. Using this toggle, we will ask 3 people to guess how much to order with the inference and uncertainty estimates and 3 other people to use the model outputs to guess how much to order. The store-product time series to use and the "correct" answer will be chosen in conjunction with the client and their forecasting domain expertise.

2.2 Comparison Table

Architecture	Owner	Overall Metric	Storyboard B Slice	Spread	Calibration	Training Cost	Inference Latency	Inspect-ability
Baseline	Gil							
Simple	Jeena							
Advanced	Vishnu							